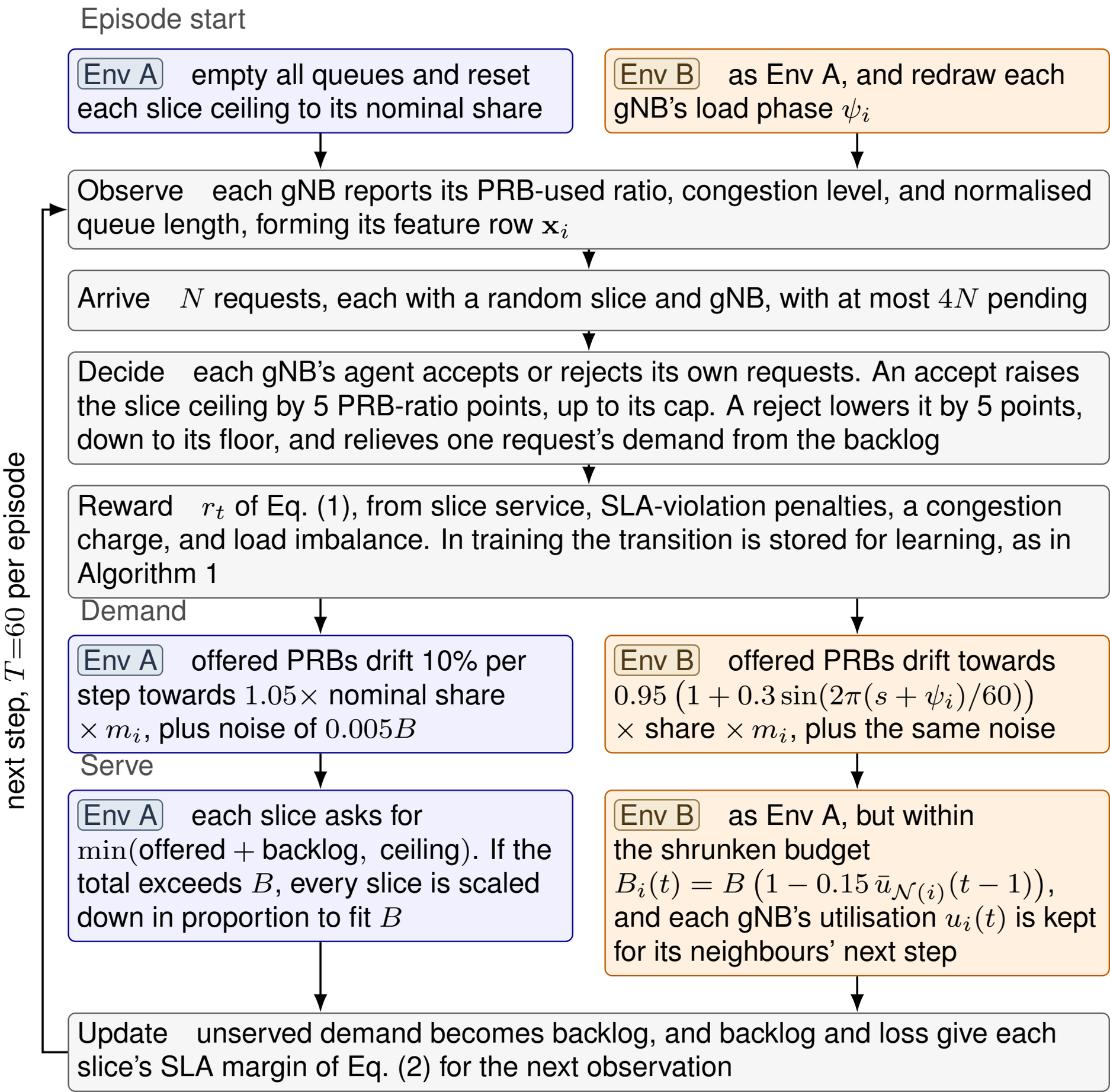

Episode start
Env A empty all queues and reset each slice ceiling to its nominal share
Env B as Env A, and redraw each gNB's load phase ψ_i
Observe each gNB reports its PRB-used ratio, congestion level, and normalised queue length, forming its feature row $\mathbf{x}_i$
Arrive N requests, each with a random slice and gNB, with at most $4N$ pending
Decide each gNB's agent accepts or rejects its own requests. An accept raises the slice ceiling by 5 PRB-ratio points, up to its cap. A reject lowers it by 5 points, down to its floor, and relieves one request's demand from the backlog
Reward r_t of Eq. (1), from slice service, SLA-violation penalties, a congestion charge, and load imbalance. In training the transition is stored for learning, as in Algorithm 1
Demand
Env A offered PRBs drift 10% per step towards $1.05\times$ nominal share $\times\, m_i$, plus noise of $0.005B$
Env B offered PRBs drift towards $0.95\,\big(1+0.3\sin(2\pi(s+\psi_i)/60)\big)$ $\times$ share $\times\, m_i$, plus the same noise
Serve
Env A each slice asks for $\min(\text{offered}+\text{backlog},\ \text{ceiling})$. If the total exceeds B, every slice is scaled down in proportion to fit B
Env B as Env A, but within the shrunken budget $B_i(t)=B\,\big(1-0.15\,\bar{u}_{\mathcal{N}(i)}(t-1)\big)$, and each gNB's utilisation $u_i(t)$ is kept for its neighbours' next step
Update unserved demand becomes backlog, and backlog and loss give each slice's SLA margin of Eq. (2) for the next observation
next step, $T{=}60$ per episode